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Malignant bowel obstruction in advanced cancer patients: epidemiology, management, and factors influencing spontaneous resolution

晚期癌症患者恶性肠梗阻：流行病学、治疗和影响自发缓解的因素

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Abstract 抽象的

Malignant bowel obstruction (MBO) is a frequent complication in advanced cancer patients, especially in those with abdominal tumors. Clinical management of MBO requires a specific and individualized approach that is based on disease prognosis and the objectives of care. The global prevalence of MBO is estimated to be 3% to 15% of cancer patients. Surgery should always be considered for patients in the initial stages of the disease with a preserved general status and a single level of occlusion. Less invasive approaches such as duodenal or colonic stenting should be considered when surgery is contraindicated in obstructions at the single level. The priority of care for inoperable and consolidated MBO is to control symptoms and promote the maximum level of comfort possible. The spontaneous resolution of an inoperable obstructive process is observed in more than one third of patients. The mean survival is of no longer than 4–5 weeks in patients with consolidated MBO. Polymodal medical treatment based on a combination of glucocorticoids, strong opioids, antiemetics, and antisecretory drugs achieves very high symptomatic control. This review focuses on the epidemiological aspects, diagnosis, surgical criteria, medical management, and factors influencing the spontaneous resolution of MBO in advanced cancer patients.

恶性肠梗阻（MBO）是晚期癌症患者的常见并发症，尤其是腹部肿瘤患者。 MBO 的临床管理需要基于疾病预后和护理目标的具体和个体化的方法。据估计，全球癌症患者中 MBO 的患病率为 3% 至 15%。对于处于疾病初始阶段且一般状态完好且单层闭塞的患者，应始终考虑手术。当单节段梗阻存在手术禁忌时，应考虑采用十二指肠或结肠支架置入术等侵入性较小的方法。对于无法手术和合并的 MBO，护理的首要任务是控制症状并尽可能提高舒适度。在超过三分之一的患者中观察到无法手术的梗阻过程会自行消退。合并 MBO 患者的平均生存期不超过 4-5 周。基于糖皮质激素、强效阿片类药物、止吐药和抗分泌药物组合的多模式医疗可实现非常高的症状控制。本综述重点关注晚期癌症患者 MBO 的流行病学方面、诊断、手术标准、医疗管理以及影响自然消退的因素。

Feedback

Keywords: malignant bowel obstruction, cancer, intestinal obstruction, bowel occlusion

【关键词】： 恶性肠梗阻 癌症 肠梗阻 肠闭塞

Introduction 介绍

Malignant bowel obstruction (MBO) is a frequent complication in patients with advanced cancer, especially of digestive or gynecological origin. Bowel obstruction is any mechanical or functional obstruction of the intestine that prevents physiological transit and digestion. This is a generic definition that is widely accepted by most authors and includes very different benign or malignant clinical situations. An international consensus group recently proposed an operative, specific definition of MBO with the aim of unifying the diagnostic criteria of this complication. According to this definition, the diagnostic criteria of MBO are: (a) clinical evidence of bowel obstruction, (b) obstruction distal to the Treitz ligament, (c) the presence of primary intra-abdominal or extra-abdominal cancer with peritoneal involvement, and (d) the absence of reasonable possibilities for a cure.[1]

恶性肠梗阻（MBO）是晚期癌症患者的常见并发症，尤其是消化道或妇科癌症患者。肠梗阻是肠道的任何机械性或功能性阻塞，阻碍生理运输和消化。这是大多数作者广泛接受的通用定义，包括非常不同的良性或恶性临床情况。一个国际共识小组最近提出了 MBO 的操作性具体定义，旨在统一该并发症的诊断标准。根据这个定义，MBO的诊断标准是：（a）肠梗阻的临床证据，（b）特雷茨韧带远端梗阻，（c）存在原发性腹内或腹外癌症并累及腹膜，(d) 不存在合理的治愈可能性。[1]

The clinical management of MBO requires a specific and individualized approach based on disease prognosis and the objectives of care. Palliative surgery – the only treatment able to restore digestive transit in consolidated MBO – must always be considered, but it should not be routinely performed. The decision making process is difficult, especially in advanced phases of cancer and depends on the level of obstruction, the presence of single or multiple occlusive levels, the extent of the cancer, associated comorbidities, and the performance status of the patient. When surgical or minimally invasive surgical approaches are not possible, a devastating clinical picture develops, which leads to intense symptoms, rapid deterioration of the patient's general status, and a short life expectancy. At this time, palliative medical treatment aimed at reducing symptoms and providing the highest level of comfort possible becomes the priority of care.

MBO 的临床管理需要根据疾病预后和护理目标采取具体的个体化方法。必须始终考虑姑息手术——唯一能够恢复合并 MBO 消化转运的治疗方法，但不应常规进行。决策过程很困难，尤其是在癌症晚期，并且取决于梗阻程度、单个或多个闭塞水平的存在、癌症的范围、相关合并症以及患者的体能状态。当无法进行外科手术或微创手术时，就会出现破坏性的临床情况，导致严重的症状、患者一般状况的迅速恶化和预期寿命的缩短。此时，旨在减轻症状并提供尽可能最高舒适度的姑息治疗成为护理的首要任务。

This review focuses on the epidemiological aspects, diagnosis, surgical criteria, medical management, and factors influencing the spontaneous resolution of MBO in advanced cancer patients.

本综述重点关注晚期癌症患者 MBO 的流行病学方面、诊断、手术标准、医疗管理以及影响自然消退的因素。

Epidemiological aspects 流行病学方面

The global prevalence of MBO is estimated to range from 3% to 15% of cancer patients, reaching 20%–50% in patients with ovarian cancer and 10%–29% in patients with colon cancer.[2–5] Primary cancers of abdominal origin that most frequently produce MBO are those of the colon (25%–40%), ovary (16%–29%), stomach (6%–19%), the pancreas (6%–13%), the bladder (3%–10%), and the endometrium (3%–11%).[2,5–9] The primary cancers of extra-abdominal origin most frequently leading to MBO due to peritoneal infiltration are those of the breast (2%–3%) and melanoma (3%).[2,5] The mean age of the patients presenting MBO is 61 years (from 58–65 years) and 64% (59%–69%) are women (Table 1). The mean time from the initial diagnosis of cancer to MBO is 14 months (13–15 months). The diagnosis of cancer coincides with the episode of MBO in 22% (13%–32%) of the cases in surgical series and in 2% in studies of patients with advanced or terminal diseases. One quarter of advanced and terminal cancer patients with this complication have presented previous episodes of intestinal obstruction (mean 1.37 subocclusive episodes per patient, SD ± 0.7).[5] The spontaneous resolution of the occlusive picture occurs in 36% (31%–42%) of patients with inoperable MBO. In these cases, the rate of recurrence of obstruction is greater than 60%. In a series of surgical cases, the average survival ranges from 3 to 8 months, including patients treated with palliative surgery. In advanced cancer patients with inoperable MBO, the mean survival rate is no

longer than 4–5 weeks. Likewise, six-month life expectancy is approximately 50% in surgical patients and 8% in patients with inoperable MBO.[5–9]

据估计，全球 MBO 患病率在癌症患者中为 3% 至 15%，在卵巢癌患者中达到 20%–50%，在结肠癌患者中达到 10%–29%。[2–5] 最常产生 MBO 的原发性腹部癌症是结肠癌 (25%–40%)、卵巢癌 (16%–29 %）、胃（6%–19%）、胰腺（6%–13%）、膀胱（3%–10%）和子宫内膜（3%–11%）。[2 5–9] 腹膜外浸润最常导致 MBO 的原发性癌症是乳腺癌(2%–3%) 和黑色素瘤 (3%)。[2 5] 出现 MBO 的患者平均年龄为 61 岁（58-65 岁），其中 64% (59%-69%) 为女性（表 1）。从癌症初次诊断到 MBO 的平均时间为 14 个月（13-15 个月）。在手术系列中，22% (13%–32%) 的病例与晚期或末期疾病患者的研究中，2% 的癌症诊断与 MBO 发作同时发生。四分之一患有这种并发症的晚期和晚期癌症患者曾出现过肠梗阻（每位患者平均有 1.37 次闭塞性发作，SD ± 0.7）。[5] 36% (31%–42%) 的无法手术的 MBO 患者会出现闭塞图像的自发消退。在这些情况下，梗阻复发率大于 60%。在一系列手术病例中，平均生存期为3至8个月，包括接受姑息性手术治疗的患者。对于无法进行 MBO 的晚期癌症患者，平均生存率不超过 4-5 周。同样，手术患者的六个月预期寿命约为 50%，无法手术的 MBO 患者的预期寿命约为 8%。[5–9]

Table 1 表格1

Characteristics and outcome of malignant bowel disease

恶性肠病的特征和结果

	Miller[6] 米勒 [6]	Blair[9] 布莱尔 [9]	Tuca[5] 图卡 [5]	Arivuex[8] 阿里维克斯 [8]
N°	32	63	100	80
Type of study 研究类型				
Surgical case series 手术病例系列	+	+	–	–
Nonsurgical series 非手术系列	–	–	+	+
Evolutive state 进化状态				
Variable 多变的	+	–	–	–
Advanced 先进的	–	+	+	+
Inoperable 无法操作	–	–	+	+
Sex (%) 性别(%)				
Women 女性	69	60	59	64
Men	31	40	41	36
Mean age (years) 平均年龄（岁）	63	58	65	64
MBO as first diagnosis of cancer (%) MBO 作为癌症的首次诊断 (%)	13	32	2	–
Time from diagnosis and MBO episode (mean in months) 从诊断到 MBO 发作的时间（平均数月）	13	15	14	–
Survival (mean in months) 生存率（平均数月）	8.5	3.0	0.8	1.2
Six-month life expectancy (%) 六个月预期寿命 (%)	50	–	8	–
Surgical resolution (%) 手术分辨率 (%)	80	37	0	0
Spontaneous resolution with conservative treatment (%) 保守治疗自然消退（%）	–	–	42	31
Rate of reobstruction after MBO resolution (%) MBO 解决后的再梗阻率 (%)	57	–	62	–

Abbreviation: MBO, malignant bowel obstruction.

缩写：MBO，恶性肠梗阻。

The higher overall frequency of MBO in women can be explained by the high incidence of this complication in ovarian cancers. The reason for the better survival observed in the surgical series is obvious; the MBO is diagnosed at an earlier stage of the disease when palliative surgery is still an option in most of the cases (80%–40%).

女性 MBO 总体发生率较高可以用卵巢癌中这种并发症的高发生率来解释。在手术系列中观察到更好的生存率的原因是显而易见的； MBO 在疾病的早期阶段被诊断出来，此时在大多数情况下仍可以选择姑息手术 (80%–40%)。

Although these data are of great interest, the current research is far from revealing the overall incidence or prevalence of this complication in cancer patients. The global context of these studies is limited since most of these data are based on selected subpopulations or retrospective case series with different outcomes and heterogeneous diagnostic criteria.

尽管这些数据非常令人感兴趣，但目前的研究还远远没有揭示癌症患者中这种并发症的总体发生率或患病率。这些研究的全球背景是有限的，因为大多数数据都是基于选定的亚群或回顾性病例系列，具有不同的结果和异质的诊断标准。

Physiopathology 病理生理学

MBO may appear at any time during the evolution of the disease, but is more frequent in cases of advanced cancer (Table 2). Obstruction may originate in the small (61%) or large bowel (33%) or in both simultaneously (20%).[10] Obstruction may be complete or partial and may appear as a subocclusive crisis or may involve one or multiple intestinal levels. In advanced and inoperable patients, multiple occlusive levels are presented in 80% of cases and peritoneal carcinomatosis is previously diagnosed in more than 65% of cases.[5] Abdominal tumor growth may lead to MBO by extrinsic intestinal compression, endoluminal obstruction, intramural infiltration, or extensive mesenteric infiltration (Figure 1). Intraluminal tumors may occlude the bowel lumen or provoke intussusception. Intramural infiltration through the mucosa may obstruct the lumen or impair peristaltic movements. Mesenteric and omental tumor involvement may angulate the bowel and provoke extramural bowel occlusion. Infiltration of the enteric or celiac plexus may cause severe impairment in peristalsis and consequent obstruction due to dysmotiliy. Factors that may favor the appearance of MBO, but are not directly dependent on abdominal tumor growth include paraneoplastic neuropathies, chronic constipation, intestinal dysfunction induced by opioids, inflammatory bowel disease, renal insufficiency, dehydration, mesenteric thrombosis, surgical adherences, and radiogenic fibrosis.

MBO 可能在疾病演变过程中的任何时间出现，但在晚期癌症病例中更为常见（表 2）。梗阻可能起源于小肠 (61%) 或大肠 (33%) 或同时起源于两者 (20%)。[10] 梗阻可能是完全的或部分的，并且可能表现为闭塞危象，或者可能涉及一个或多个肠道水平。在晚期和无法手术的患者中，80% 的病例出现多个闭塞水平，并且超过 65% 的病例先前被诊断出腹膜癌病。[5] 腹部肿瘤生长可能因外源性肠道压迫、腔内梗阻、壁内浸润或广泛肠系膜浸润而导致 MBO（图 1）。腔内肿瘤可能会堵塞肠腔或引起肠套叠。通过粘膜的壁内渗透可能会阻塞管腔或损害蠕动运动。肠系膜和大网膜肿瘤的受累可能会使肠道成角并引起肠壁外闭塞。肠或腹腔神经丛的浸润可能会导致蠕动严重受损，并因运动障碍而导致梗阻。可能有利于 MBO 出现但不直接依赖于腹部肿瘤生长的因素包括副肿瘤性神经病、慢性便秘、阿片类药物引起的肠道功能障碍、炎症性肠病、肾功能不全、脱水、肠系膜血栓形成、手术依从性和放射性纤维化。

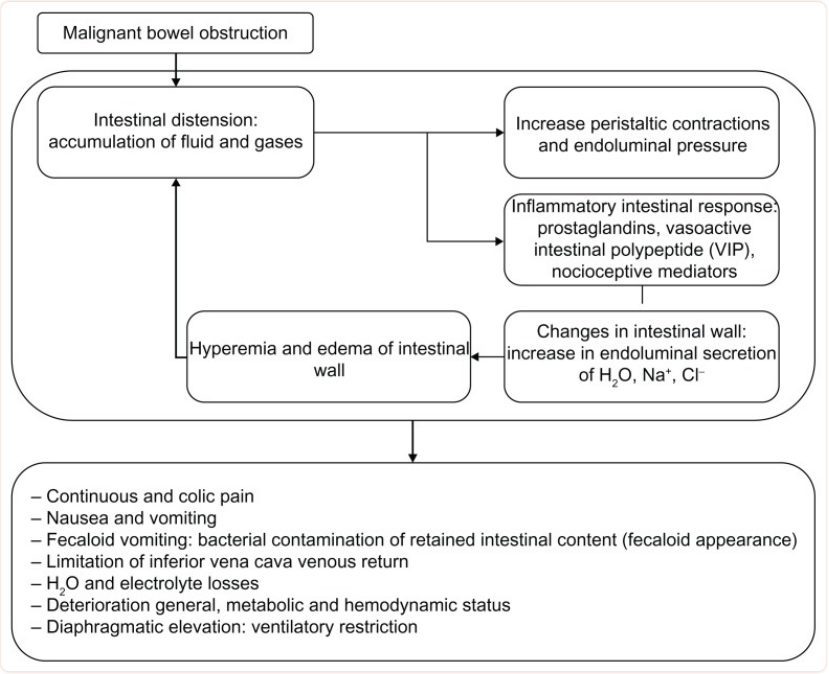


Figure 1 图1

Physiopathology of MBO. MBO 的病理生理学。

Table 2 表2

Physiopathology of MBO (I)

MBO的病理生理学（一）

Factors directly related to intra-abdominal tumor growth
与腹腔内肿瘤生长直接相关的因素
Extrinsic intestinal compression
外源性肠道压迫
Endoluminal intestinal obstruction
腔内肠梗阻
Intramural intestinal infiltration
肠壁间浸润
Infiltration of the mesenterium and plexus
肠系膜和神经丛的浸润
Factors not directly related to intra-abdominal tumor growth
与腹腔内肿瘤生长不直接相关的因素
Paraneoplastric neuropathy
肿瘤旁神经病
Chronic constipation 慢性便秘
Opioid-induced intestinal dysfunction
阿片类药物引起的肠道功能障碍
Adynamic ileum 回肠动力不足
Inflammatory intestinal disease
炎症性肠道疾病
Renal insufficiency/dehydratation
肾功能不全/脱水
Mesenteric thrombosis 肠系膜血栓形成
Postsurgical adherences 术后依从性
Radiogenic fibrosis 放射源性纤维化

Fluid retention and intestinal gases proximal to the occlusive level produce a marked increase in endoluminal intestinal pressure. This abdominal distension favors the release of 5-HT_3 by the intestinal enterochromaffin cells which, in turn, activates the enteric interneuronal system through its different mediators (P substance, nitric oxide, acetylcholin, somatostatin, and vasoactive intestinal peptide). This stimulates the secretomotor neurons that are especially mediated by the vasoactive intestinal peptide, which leads to splanchnic vasodilatation and hypersecretion of the cells of the intestinal crypts. The consequences of these phenomena are the appearance of intense intestinal edema, an increase in the secretions retained, and endoluminal pressure, all of which are mechanisms that perpetuate the physiopathological process of MBO.[2,7,11–13]

接近闭塞水平的液体潴留和肠道气体导致肠腔内压力显着增加。这种腹胀有利于肠道肠嗜铬细胞释放 5-HT_3，进而通过其不同的介质（P 物质、一氧化氮、乙酰胆碱、生长抑素和血管活性肠激素）激活肠内神经元系统。肽）。这刺激了尤其由血管活性肠肽介导的分泌运动神经元，导致内脏血管舒张和肠隐窝细胞分泌过多。这些现象的后果是出现严重的肠水肿、滞留分泌物增加和腔内压力增加，所有这些都是使 MBO 的病理生理过程持续下去的机制。[2 7 11 – 13]

Clinical manifestations 临床表现

The onset of MBO may be subacute with the presence of colic pain, abdominal distension, nausea, and vomiting, which spontaneously cease (subocclusive crisis). The prevalence of symptoms in consolidated MBO are nausea 100%, vomiting 87%–100%, colic pain 72%–80%, pain due to distension 56%–90%, and the absence of stools or emission of gases in the previous 72 hours in 85%–93%.[2,5,7,8] In upper MBO, the nausea is intense and presents early, the vomiting is numerous and with an aqueous, mucous or biliary appearance and has little odor. Vomiting in lower obstruction usually occurs later, is dark, and has a strong odor. Bacterial liquefaction of the retained intestinal content in the zone proximal to the obstruction confers the characteristic appearance and smell of fecaloid vomit. Cases with partial obstruction may present liquid stools due to bacterial liquefaction of the digestive content and intestinal hypersecretion. The colic pain is due to giant peristaltic waves and spasms in the bowel with increased endoluminal pressure and no possibility of effective transit. Intestinal distension and tumoral infiltration of the abdominal structures are responsible for the continuous pain.[2,10,14] During physical examination, abdominal distension is noted and is more marked in lower obstructions and in the changes in peristaltism. At the onset, borborygmus, fighting peristalsis, may be heard on auscultation. On consolidation of MBO, the peristalsis may decrease or even cease to present isolated metallic sounds thereafter due to hydroaerial tension on auscultation. In patients with advanced cancer, MBO is also associated with anemia (70%), hypoalbuminemia (68%), alterations in hepatic enzymes (62%), dehydration and prerenal renal dysfunction (44%), cachexia (22%), ascites (41%), palpable abdominal tumor masses (21%), and marked cognitive deterioration (23%).[5]

MBO 的发作可能是亚急性的，伴有绞痛、腹胀、恶心和呕吐，但这些症状会自行停止（亚闭塞危象）。综合 MBO 症状的发生率为恶心 100%、呕吐 87%–100%、绞痛 72%–80%、腹胀疼痛 56%–90%、以及过去 72 天内没有大便或排出气体 85%–93% 的小时数。[2 5 7 8] 在上位 MBO 中，恶心剧烈且较早出现，呕吐次数多且伴有水样粘液或胆汁外观，几乎没有气味。下位梗阻的呕吐通常发生较晚，呕吐物呈黑色且有强烈的气味。靠近梗阻区域残留的肠内容物的细菌液化赋予了粪便样呕吐物的特征性外观和气味。部分梗阻的病例可能因消化内容物细菌液化和肠道分泌过多而出现液状粪便。绞痛是由于肠内巨大的蠕动波和痉挛导致肠腔内压力增加并且不可能有效传输。肠扩张和腹部结构的肿瘤浸润是造成持续疼痛的原因。[2 10 14] 体检时发现腹胀，下部梗阻和蠕动变化更明显。发病时，听诊时可能会听到肠鸣，对抗蠕动。在 MBO 巩固时，由于听诊时的空气压力，蠕动可能会减少，甚至停止出现孤立的金属声音。 在晚期癌症患者中，MBO 还与贫血（70%）、低蛋白血症（68%）、肝酶改变（62%）、脱水和肾前性肾功能不全（44%）、恶病质（22%）、腹水（41%），可触及腹部肿瘤肿块（21%），以及明显的认知能力下降（23%）。[5]

Radiological diagnosis 放射诊断

Plain radiography of the abdomen in a biped position is the imaging method of choice for the detection of suspected MBO and is also used to assess the patient's evolution after treatment. The radiological signs of MBO are distension of the intestinal loops, fluid retention, and gases with the presence of air-fluid levels in the zone proximal to the occlusion as well as a reduction in gas and stools in the segments distal to the obstruction. In upper occlusions, distension of the loops and air-fluid levels may be absent. Radiological techniques using contrast may be necessary to evaluate the surgical approach. Barium contrast provides excellent radiological definition, but is not absorbed and may become impacted, thereby compromising other tests or endoscopic maneuvers. In many cases, these imaging tests are limited by the presence of nausea and vomiting, which may prevent the ingestion of radiographic contrast or increase the risk of aspiration pneumonia.

双足位置腹部平片是检测可疑 MBO 的首选成像方法，也可用于评估患者治疗后的进展。 MBO 的放射学症状是肠袢扩张、液体潴留和气体，靠近闭塞的区域存在气液水平，以及闭塞远端区域气体和粪便减少。在上部闭塞中，循环扩张和气液水平可能不存在。可能需要使用造影剂的放射学技术来评估手术方法。钡造影剂可提供出色的放射清晰度，但不会被吸收并可能受到影响，从而影响其他测试或内窥镜操作。在许多情况下，这些影像学检查受到恶心和呕吐的限制，这可能会妨碍放射造影剂的摄入或增加吸入性肺炎的风险。

Gastrografin provides similar radiologic definition and its hyperosmotic character may, in some cases, favor the resolution of obstructions in the small bowel. In fact, a recent meta-analysis confirms a reduction in the need for surgical intervention and hospital stay in patients with occlusion after the administration of Gastrografin.[15] Computerized tomography (CT) provides a high possibility for the diagnosis of the extension of the neoplasm and, on many occasions, the level of obstruction. The diagnostic sensitivity of CT in determining the obstruction level is of 93%, with a specificity of 100% and a predictive value of 83%–94%, which is significantly higher than that provided by abdominal echography and simple radiology.[16,17] The precision of the diagnosis of peritoneal carcinomatosis by CT is scarce, with a predictive value of less than 20% if the peritoneal lesions are less than 0.5 cm or if they are located in the pelvis, mesenterium, or small bowel.[18,19] The sensitivity of magnetic resonance (MR) in diagnosing of the extension of the neoplasm and the level of the obstruction is 93%–95%, with a specificity of 63%–100% and a predictive value of 81%–96%.

One study on the diagnostic possibilities of MR compared with CT in MBO, showed the significant superiority of MR in terms of sensitivity, specificity, and predictive value.[21]

泛影葡胺提供类似的放射学定义，其高渗特性在某些情况下可能有利于小肠梗阻的解决。事实上，最近的一项荟萃分析证实，服用泛影葡胺后，闭塞患者的手术干预和住院时间减少了。[15] 计算机断层扫描 (CT) 为诊断肿瘤的扩展以及在许多情况下的阻塞程度提供了很高的可能性。 CT确定梗阻程度的诊断敏感性为93%，特异性为100%，预测价值为83%~94%，明显高于腹部超声检查和单纯放射学检查。[16,17] CT诊断腹膜癌病的精确度较低，腹膜病灶小于0.5cm或位于腹膜癌病灶的预测值低于20%在骨盆、肠系膜或小肠中。[18,19] 磁共振 (MR) 诊断肿瘤扩展和阻塞程度的敏感性为 93%–95%，特异性为 63%– 100%，预测值为 81%–96%。一项关于 MR 与 CT 诊断 MBO 可能性的研究表明，MR 在敏感性、特异性和预测价值方面具有显着优势。[21]

In summary: 总之：

– Plain abdominal radiography is sufficient in most cases to confirm the diagnosis of MBO.

– 在大多数情况下，腹部平片足以确诊 MBO。

– One should consider the use of contrast radiography, CT, or MR, when the patient's general condition was good prior to the complication, the extent of the cancer is unknown, when a unique occlusive level is suspected, and when the cancer is potentially operable.

– 当患者在出现并发症之前一般状况良好、癌症范围未知、怀疑存在独特的闭塞水平以及癌症可能可以手术时，应考虑使用造影剂、CT 或 MR 。

– Contrast radiography determines, with a reasonable degree of accuracy, the site or sites of obstruction and the degree of obstruction. It may rule out a bowel occlusion due to motility disorder (opioid-induced intestinal dysfunction, pseudo-obstruction).

– 造影射线照相以合理的准确度确定梗阻部位和梗阻程度。它可以排除由于动力障碍（阿片类药物引起的肠道功能障碍，假性梗阻）引起的肠闭塞。

– CT or MR should be reserved for cases where precise radiological information is needed to facilitate adequate decision making regarding surgery (ie, tumor characteristics in the site of obstruction, the presence of lymph nodes, and the intra- and extra-abdominal metastatic spread).

– CT 或 MR 应保留用于需要精确放射学信息以促进做出适当手术决策的情况（即梗阻部位的肿瘤特征、淋巴结的存在以及腹内和腹外转移扩散） 。

Treatment 治疗

The decision making process in advanced oncologic patients requires individualized evaluation based on the extension of the neoplasm, the global prognosis, the possibility of specific cancer treatments, associated comorbidities, the general status, and the particular options available to the duly informed patient. Possible treatments include surgery, endoscopic palliation, digestive aspiration, and symptomatic palliative pharmacologic therapy.

晚期肿瘤患者的决策过程需要根据肿瘤的范围、总体预后、特定癌症治疗的可能性、相关合并症、一般状况以及适当知情的患者可用的特定选择进行个体化评估。可能的治疗方法包括手术、内窥镜姑息治疗、消化道抽吸和对症姑息药物治疗。

Surgery in MBO MBO 手术

The aim of surgery is to reestablish digestive permeability, and should always be considered in patients in the initial stages of the disease, with a preserved general status and a single level of occlusion. Studies involving a series of surgical cases of MBO have shown a 30-day mortality of 25% (9%–40%), postsurgical morbidity of 50% (9%–90%), a rate of reobstruction of 48% (39%–57%), and a median survival of 7 months (2–12 months).[6,9,10,23–28]

手术的目的是重建消化道通透性，对于处于疾病初始阶段、保留一般状态和单级闭塞的患者，应始终考虑手术。涉及一系列 MBO 手术病例的研究表明，30 天死亡率为 25% (9%–40%)，术后发病率为 50% (9%–90%)，再梗阻率为 48% (39%) – 57%），中位生存期为 7 个月（2-12 个月）。[6,9,10,23–28]

Age, advanced disease, malnutrition, and deterioration in the general status are considered factors of poor prognosis even in cases where surgery may technically be possible.[10,14] A study on patients with colon cancer undergoing surgery for MBO reported an increase in surgical mortality associated with age, with an OR of 1.85 for each 10-year interval of age above 65 years. Using the American Society of Anesthesia scale to measure deterioration of general status, surgical mortality (Odds Ratio 3.3) increased in patients with a score ≥ 2, compared to those with a score < 2.[29] Furthermore, surgical mortality is three-fold greater in patients

with deficient nutritional status and hypoalbuminemia.[9,30] The presence of ascites greater than 3000 mL and palpable tumor masses is statistically associated with a poor surgical prognosis.[9,30]

即使在技术上可能进行手术的情况下，年龄、晚期疾病、营养不良和一般状况恶化也被认为是预后不良的因素。[10 14] 一项针对接受 MBO 手术的结肠癌患者的研究报告称，手术死亡率随年龄增加而增加，年龄在 65 岁以上每 10 年间隔的 OR 为 1.85年。使用美国麻醉学会量表来衡量一般状态恶化情况，与评分 < 2 的患者相比，评分 ≥ 2 的患者的手术死亡率（比值比 3.3）增加。[29] 此外，手术死亡率营养不良和低蛋白血症的患者的这一比例要高出三倍。[9 30] 腹水大于 3000 mL 且可触及肿瘤块与手术预后不良在统计学上相关。[9 30]

The mean survival for patients with ovarian cancer undergoing surgery for MBO is significantly higher in those without ascites and previous chemotherapy.[31–33] Pelvic and abdominal radiotherapy prior to MBO is associated with a high rate of surgical complications and an increase in operative mortality in patients with gynecological cancer, a fact that has not been confirmed in cancer of other etiologies.[9,32] Medical treatment prior to surgery for MBO is based on absolute diet, parenteral hydration, nasogastric aspiration, and antiemetic and analgesic drugs. The aims of these measures are to control the symptoms, reestablish the hydroelectrolytic balance, favor spontaneous resolution, and gain the time necessary to establish a diagnostic process to facilitate individualized surgical decisions. With these measures, adequate control of the symptoms is achieved in 80% of cases if absolute diet and nasogastric aspiration are maintained. It is reasonable to assume that nasogastric aspiration at the onset of the obstruction may favor spontaneous resolution since it drastically reduces endoluminal pressure. However, longterm nasogastric aspiration is uncomfortable for the patient and has intense secondary effects (eg, esophagitis, gastroesophageal reflux, nasal erosions, and bronchoaspiration). In a series of surgical cases, spontaneous resolution is presented in 30% of patients within a mean time of less than 8 days after diagnosis. For this reason, and considering the hypothesis that nasogastric aspiration may improve the rate of spontaneous resolution, there is no reason to maintain this measure over a prolonged period of time greater than the possible estimations of resolution in patients in whom surgery has been ruled out and in whom adequate control of the symptomatology may be obtained with intensification of palliative treatment.

接受 MBO 手术的卵巢癌患者的平均生存期明显高于没有腹水且既往接受过化疗的患者。[31 – 33] MBO 之前的盆腔和腹部放疗与妇科癌症患者的高手术并发症发生率和手术死亡率增加相关，这一事实尚未在其他病因的癌症中得到证实。[9 32] MBO 手术前的药物治疗基于绝对饮食、肠外补液、鼻胃抽吸以及止吐和镇痛药物。这些措施的目的是控制症状，重建水电解平衡，促进自发消退，并获得建立诊断过程所需的时间，以促进个体化手术决策。通过这些措施，如果保持绝对饮食和鼻胃抽吸，80% 的病例可以充分控制症状。可以合理地假设，梗阻开始时进行鼻胃抽吸可能有利于自发缓解，因为它大大降低了腔内压力。然而，长期鼻胃抽吸会给患者带来不适，并产生严重的副作用（例如食管炎、胃食管反流、鼻糜烂和支气管误吸）。在一系列手术病例中，30% 的患者在诊断后平均不到 8 天的时间内自行痊愈。 因此，考虑到鼻胃抽吸可能会提高自发消退率的假设，没有理由在较长时间内维持这一措施，而不是超过已排除手术的患者可能的消退估计值。通过加强姑息治疗可以充分控制症状。

Taking these data into account, most researchers consider the following factors to limit the indication of surgery in MBO: elderly age, malnutrition or cachexia, peritoneal carcinomatosis, multiple occlusive levels, palpable abdominal masses, previous malnutrition, refractory ascites, symptomatic extra-abdominal metastatic disease, deteriorated general status, renal or hepatic insufficiency, previous abdominal pelvic radiotherapy, and the absence of the possibility for specific oncologic treatments.[2,10]

考虑到这些数据，大多数研究人员认为以下因素限制了MBO的手术指征：高龄、营养不良或恶病质、腹膜癌病、多发性闭塞水平、可触及的腹部肿块、既往营养不良、难治性腹水、有症状的腹外转移疾病、全身状况恶化、肾或肝功能不全、既往接受过腹部盆腔放疗，以及无法进行特定的肿瘤治疗。[2 10]

Endoscopic palliation and stents

内窥镜姑息治疗和支架

The use of stents has spread in recent years as an endoscopic alternative for the treatment of proximal small bowel obstructions and the colon. The rate of symptomatic control and tolerance of ingestion after the placement of a duodenal stent for duodenal obstruction is 90%. Studies comparing surgical gastrojejuostomy with the placement of a duodenal stent showed no significant differences in technical success (100% versus 90%), in early (6% versus 7%) or late complications (17% versus 18%), and in persistent occlusive symptomatology (9% versus 8%), respectively. The mean survival is 105 days for the group treated with a duodenal stent compared to 165 days in the surgical group.[34] The most frequent late complications after duodenal stent placement are hemorrhage or perforation (1.2%), migration of the device (5%), and stent obstruction (18%),[35] which is sometimes susceptible to repermeabilization using a ND Yang laser.[36] Successful insertion of a stent in colon cancer ranges from 80%–100% of the cases and improves the symptomatology in more than 75% of the patients. The mean duration of colonic stent permeability is 106 days (66–88 days). The most frequent complications of this technique are immediate or differed perforation (4.5%), migration (11%), and obstruction (12%).[37,38]

Recurrence of obstruction due to tumor growth through the mesh or endoluminal at the ends of the stent may require the placement of a second additional stent after repermeabilizing the lumen with a laser or photodynamic therapy.

近年来，支架作为治疗近端小肠梗阻和结肠的内窥镜替代方案已得到广泛应用。十二指肠梗阻置入十二指肠支架后症状控制率和摄食耐受率为90%。比较胃空肠造口术与十二指肠支架置入术的研究表明，在技术成功率（100% 对 90%）、早期并发症（6% 对 7%）或晚期并发症（17% 对 18%）以及持续闭塞方面没有显着差异。症状学（分别为 9% 和 8%）。十二指肠支架治疗组的平均生存期为 105 天，而手术组的平均生存期为 165 天。[34] 十二指肠支架置入后最常见的晚期并发症是出血或穿孔（1.2%）、装置移位（5%）和支架阻塞（18%），[35] 有时容易受到 ND 杨激光再透化的影响。[36] 结肠癌支架的成功置入率为 80%–100%，并且改善了超过 75% 的患者的症状。结肠支架通透性的平均持续时间为 106 天（66-88 天）。该技术最常见的并发症是直接或不同程度的穿孔（4.5%）、移位（11%）和阻塞（12%）。[37] [38] 由于肿瘤通过支架末端的网状或腔内生长而再次发生阻塞，可能需要在用激光或光动力对管腔进行再透化后放置第二个附加支架治疗。

In summary, self-expanding metal stents can be considered a good option in patients with a single point of obstruction in whom palliative surgery has been ruled out or in those who do not want to undergo surgery.

综上所述，对于已排除姑息性手术或不想接受手术的单点梗阻患者，自膨式金属支架不失为一个不错的选择。

Percutaneous endoscopic gastrostomy by aspiration

经皮内镜下胃造口术

As mentioned previously, longterm aspiration using a nasogastric tube is uncomfortable and may produce severe secondary effects. Percutaneous endoscopic gastrostomy may be a highly effective and safe alternative for patients in whom surgery is ruled out and in those who cannot control their symptoms without maintaining digestive aspiration. In MBO secondary to advanced ovarian cancer, the success of percutaneous endoscopic placement is 94% (94%–98%), which achieves adequate control of symptomatology in 84% of the patients during a mean time of 70 days, even in cases presenting peritoneal carcinomatosis, ascites, or gastric infiltration.[39,40]

如前所述，长期使用鼻胃管抽吸会令人不舒服，并可能产生严重的副作用。对于那些无法接受手术的患者以及那些在不维持消化道抽吸的情况下无法控制症状的患者，经皮内窥镜胃造口术可能是一种高效且安全的替代方案。在继发于晚期卵巢癌的 MBO 中，经皮内镜置入术的成功率为 94% (94%–98%)，即使在出现腹膜癌的情况下，84% 的患者在平均 70 天的时间内实现了症状的充分控制癌病、腹水或胃浸润。[39] [40]

Parenteral nutrition in MBO

MBO 中的肠外营养

The aim of total parenteral nutrition (TPN) is the recovery of nutritional status in patients who are candidates for surgery. The indication for TPN in advanced oncological patients with inoperable MBO is more controversial. TPN is an invasive technique that requires specific training for use and frequent monitoring of electrolytes and hydration. It also predisposes patients to infection (central venous access), thrombosis, diarrhea, liver dysfunction, and hyperglycemia. The scarce studies that have evaluated the efficacy of longterm TPN for inoperable MBO have reported a mean survival rate of 4–6 months, a rate of complications associated with the procedure greater than 13%, and maintained stability of nutritional parameters of only 2–3 months prior to death. These studies concluded that only 30% of the patients surviving for more than 3 months, benefit from the application of TPN.[41,42] Routine use of TPN in MBO is not recommended in inoperable patients. The indication for long term TPN should be made with caution and should be reserved for patients with a preserved general status prior to MBO, slow growing tumors, the possibility of response to chemotherapy, reasonable expectations of survival longer than 3 months, and without severe extra-abdominal complications by the extension of the neoplasm.

全肠外营养（TPN）的目的是恢复手术患者的营养状况。对于无法手术的 MBO 晚期肿瘤患者进行 TPN 的适应症更具争议性。 TPN 是一种侵入性技术，需要进行特定的使用培训并经常监测电解质和水合作用。它还使患者容易感染（中心静脉通路）、血栓形成、腹泻、肝功能障碍和高血糖。评估长期 TPN 对无法手术的 MBO 疗效的研究很少，报告称平均生存率为 4-6 个月，与手术相关的并发症发生率大于 13%，并且营养参数的稳定性仅维持 2-3 个月。死亡前几个月。这些研究得出的结论是，只有 30% 存活超过 3 个月的患者从 TPN 的应用中受益。[41] [42] 不建议无法手术的患者在 MBO 中常规使用 TPN。应谨慎制定长期 TPN 的指征，并应保留给 MBO 前一般状态良好、肿瘤生长缓慢、对化疗有反应的可能性、合理预期生存时间超过 3 个月、且无严重额外症状的患者。 -肿瘤扩展引起的腹部并发症。

Palliative treatment 姑息治疗

In 1985, Baines et al demonstrated that pharmacologic treatment, specifically palliative treatment for inoperable MBO, may provide adequate symptomatic control with measures aimed at maintaining the maximum comfort possible.[4] Palliative treatment for MBO has the following objectives: control of nausea, vomiting, and pain, allowing minimum food intake, avoiding or withdrawing nasogastric aspiration, and favoring hospital discharge under control at home or in healthcare centers. This treatment is based on the use of antiemetic, potent analgesic, glucocorticoid, and antisecretor drugs in combination with the most comfortable route of administration to allow its application within the homecare setting.[2,4]

1985 年，Baines 等人证明，药物治疗，特别是无法手术的 MBO 的姑息治疗，可以通过旨在保持最大舒适度的措施提供充分的症状控制。[4] MBO 的姑息治疗有以下目标：控制恶心、呕吐和疼痛，允许最低限度的食物摄入，避免或停止鼻胃抽吸，并有利于在家中或医疗中心控制出院。这种治疗基于使用止吐药、强效镇痛药、糖皮质激素和抗分泌药物，并结合最舒适的给药途径，使其能够在家庭护理环境中应用。[2,4]

More than 80% of patients with MBO present continuous pain and high-intensity colic.[4,5,7,8] The administration of analgesics for the treatment of MBO should be adjusted to the analgesic scale of the World Health Organization (WHO), which has demonstrated an efficacy rate greater than 80% in cancer patients.[43–46] According to the European Society of Palliative Care and the WHO, morphine is the opioid of choice in the absence of controlled clinical trials comparing different opioids in this indication.[47] Some authors have reported that oxycodone may be more effective than other opioids for visceral pain treatment given its action on the kappa opioid receptors, although this has yet to be confirmed in controlled clinical studies.[48] A meta-analysis of five controlled clinical trials by Tassinari et al in 2009 confirmed that fentanyl is a potent opioid that produces less constipation as a secondary effect.[49] A recent descriptive analysis of MBO in advanced cancer shows that more than 60% of the patients were treated with potent opioids prior to the occlusive episode and more than 80% required these drugs for analgesia during the episode. In this study, no statistically significant differences were observed in the rate of spontaneous resolution under symptomatic treatment among patients treated with a potent opioid prior to or during the episode of MBO versus those who did not receive this type of drug.[5] The opioid dose should be titrated individually for adequate pain relief. The subcutaneous, intravenous, sublingual, or transdermal route for opioid administration should be used frequently because nausea and vomiting do not allow for oral administration.

超过80%的MBO患者出现持续疼痛和高强度绞痛。[4,5,7,8] 治疗MBO时镇痛药的使用应符合世界卫生组织的镇痛标准（WHO），已证明对癌症患者的有效率超过 80%。[43–46] 根据欧洲姑息治疗协会和世界卫生组织的说法，在缺乏比较不同阿片类药物的对照临床试验的情况下，吗啡是首选阿片类药物这个指示。[47] 一些作者报告说，考虑到羟考酮对 kappa 阿片受体的作用，羟考酮治疗内脏疼痛可能比其他阿片类药物更有效，尽管这一点尚未在对照临床研究中得到证实。[48] Tassinari 等人于 2009 年对五项对照临床试验进行的荟萃分析证实，芬太尼是一种强效阿片类药物，具有减少便秘的副作用。[49] 最近对晚期癌症 MBO 的描述性分析表明，超过 60% 的患者在闭塞发作前接受了强效阿片类药物治疗，超过 80% 的患者在闭塞发作期间需要这些药物进行镇痛。在这项研究中，在 MBO 发作之前或期间接受强效阿片类药物治疗的患者与未接受此类药物的患者相比，在对症治疗下的自然缓解率没有观察到统计学上的显着差异。[5] 阿片类药物剂量应单独调整，以充分缓解疼痛。应经常使用皮下、静脉内、舌下或透皮途径阿片类药物给药，因为恶心和呕吐不允许口服给药。

Antiemetic treatment uses drugs from three pharmacological groups: anti-cholinergic, dopamine antagonists, and serotonin antagonists (5-HT_3). The dopamine antagonists are divided into benzamides (metoclopramide), butyrophenones (haloperidol), and phenothiazines (chlorpromazine). Metoclopramide blocks the dopamine receptors (D2) at the central and peripheral level. Its action facilitates the release of acetylcholine and at high doses (>120 mg/day) antagonizes the 5-HT_3 receptors. The mixed, central, and peripheral actions confer an antiemetic and prokinetic digestive effect on metoclopramide. The usual metoclopramide doses range from 40–120 mg/day. Haloperidol and phenothiazines (chlorpromazine and levomepromazine) are neuroleptic drugs that block the dopamine receptors at the central level only. They have a potent antiemetic, but not prokinetic, action. Among these drugs, haloperidol is considered the most ideal because it produces less somnolence and anticholinergic effects. Haloperidol doses range from 5–15 mg/day, which can be administered in divided doses subcutaneously or intravenously, or by continuous subcutaneous or intravenous infusion. Scopolamine and hyoscine butylbromide are anticholinergic drugs that have an antiemetic action that acts by blocking acetylcholine at the central and peripheral levels, and the peripheral level, respectively, in association with a clear antisecretor effect. Hyoscine butylbromide doses range from 40–120 mg/day. The serotonin (5-HT_3) receptor antagonists, as ondansetron or granisetron, can be useful for emesis control in the treatment of MBO. A recent noncontrolled Phase II study demonstrated an index greater than 80% for the antiemetic control of MBO using granisetron (5-HT_3 receptor antagonist), even in cases that have not responded to typical antiemetic treatment.[7] The ondansetron dose ranges from 12–24 mg/day, and the granisetron dose ranges from

1–3 mg/day. These drugs are usually well tolerated. Headache, dizziness, and constipation are the most commonly reported side effects associated with their use.

止吐治疗使用来自三个药理学组的药物：抗胆碱能药、多巴胺拮抗剂和血清素拮抗剂（5-HT $_3$ ）。多巴胺拮抗剂分为苯甲酰胺类（甲氧氯普胺）、丁酰苯类（氟哌啶醇）和吩噻嗪类（氯丙嗪）。胃复安可阻断中枢和外周水平的多巴胺受体 (D2)。其作用促进乙酰胆碱的释放，高剂量（> 120 毫克/天）可拮抗 5-HT $_3$ 受体。混合、中枢和外周作用使甲氧氯普胺具有止吐和促消化作用。甲氧氯普胺的常用剂量范围为 40-120 毫克/天。氟哌啶醇和吩噻嗪（氯丙嗪和左美丙嗪）是仅在中枢水平阻断多巴胺受体的精神安定药物。它们具有有效的止吐作用，但不具有促动力作用。在这些药物中，氟哌啶醇被认为是最理想的，因为它产生较少的嗜睡和抗胆碱能作用。氟哌啶醇剂量范围为5-15 mg/天，可分剂量皮下或静脉注射，或连续皮下或静脉输注。东莨菪碱和丁溴东莨菪碱是抗胆碱能药物，具有止吐作用，通过分别在中枢和外周水平以及外周水平阻断乙酰胆碱来发挥作用，并具有明显的抗分泌作用。丁溴东莨菪碱的剂量范围为 40-120 毫克/天。 5-羟色胺 (5-HT $_3$) 受体拮抗剂，如昂丹司琼或格拉司琼，可用于 MBO 治疗中的呕吐控制。 最近的一项非对照 II 期研究表明，即使在对典型止吐治疗没有反应的病例中，使用格拉司琼（5-HT $_3$ 受体拮抗剂）对 MBO 的止吐控制指数也大于 80%。 [7] 昂丹司琼剂量范围为12-24毫克/天，格拉司琼剂量范围为1-3毫克/天。这些药物通常耐受性良好。头痛、头晕和便秘是最常见的与其使用相关的副作用。

Glucocorticoids possess an antiemetic action, the mechanism of which is not well known, and an anti-inflammatory action that reduces peritumoral edema. Therefore, most researchers recommend glucocorticoids in the palliative treatment of MBO. A meta-analysis of three controlled clinical trials published in 1999 demonstrate that the use of glucocorticoids, particularly dexamethasone at a dose ranging from 6–16 mg, collaborates with the antiemetic action and favors the spontaneous resolution of MBO in advanced gynecological and digestive cancer. In this meta-analysis, the rate of spontaneous resolution was 62%–68% in patients treated with glucocorticoids compared to 33%–57% in those receiving placebo.[32,50,51]

糖皮质激素具有止吐作用（其机制尚不清楚）和减少瘤周水肿的抗炎作用。因此，大多数研究者推荐糖皮质激素用于MBO的姑息治疗。 1999 年发表的三项对照临床试验的荟萃分析表明，使用糖皮质激素，尤其是 6-16 mg 剂量的地塞米松，可与止吐作用协同作用，有利于晚期妇科和消化道癌症 MBO 的自发消退。在这项荟萃分析中，接受糖皮质激素治疗的患者的自发缓解率为 62%–68%，而接受安慰剂治疗的患者的自发缓解率为 33%–57%。 [32] [50] [51]

The objective of antisecretor drugs is to reduce intestinal hypersecretion, and secondarily, to improve nausea, vomiting, and pain. Anticholinergic drugs (scopolamine, hyoscine, and butylbromide) have traditionally been the antisecretors of choice. Octreotide, a somatostatin analog, provides a more specific and prolonged antisecretor effect. The pharmacologic activity of octreotide is mediated by the inhibition of the secretion of vasoactive intestinal peptides. This pharmacologic activity reduces hydroelectrolytic retention in the intestinal lumen, as well as gastric secretions, intestinal motility, biliary flow, splanchnic hypervascularization, and intestinal parietal edema. Furthermore, it increases the absorption of water and the production of intestinal mucous.[52,53] Different studies on the effectiveness of octreotide at doses from 200–600 µg/day have shown a clear reduction in intestinal secretions, the eventual possibility of withdrawing the nasogastric tube, and a high grade of antiemetic and analgesic response with no relevant adverse effects.[2,8,54–58] Two controlled clinical studies have compared the antiemetic, analgesic, and antisecretory efficacy of octreotide (300 µg/day) versus hyoscine butylbromide (60 mg/day) in the treatment of MBO. In both studies, the efficacy of octreotide was statistically greater in all the parameters of response (reduction in digestive hypersecretion and control of nausea and vomiting). [54–56] A Phase II study demonstrated that a long-acting formulation of octreotide (LAR Depot) in combination with corticosteroids is useful and safe for the treatment of MBO due to peritoneal carcinomatosis.[59] A recent review of the literature concludes that despite the limited number of controlled clinical trials, octreotide is the antisecretory agent of choice for the treatment of MBO based on the results from 15 consistent studies and the experience acquired from 20 years of its use.[60] Histamine-2 antagonists and proton pump inhibitors are useful for reducing gastric secretions. A recent meta-analysis confirmed that ranitidine is more effective than proton pump inhibitors as an antisecretory agent. Based on these data, the authors hypothesized that ranitidine would be useful as an adjuvant drug in antisecretory therapy for the treatment of MBO and suggested the development of specific research to confirm these findings.[61,62]

抗分泌药物的目的是减少肠道分泌过多，其次是改善恶心、呕吐和疼痛。抗胆碱能药物（东莨菪碱、东莨菪碱和丁基溴）历来是首选的抗分泌剂。奥曲肽是一种生长抑素类似物，具有更特异和更持久的抗分泌作用。奥曲肽的药理活性是通过抑制血管活性肠肽的分泌来介导的。这种药理活性可减少肠腔内的水电解滞留，以及胃分泌、肠蠕动、胆汁流量、内脏血管过多和肠壁水肿。此外，它还能增加水的吸收和肠粘液的产生。 [52] [53] 关于奥曲肽 200-600 µg/天剂量的有效性的不同研究表明，肠道分泌物明显减少，最终可能拔除鼻胃管，并且高度止吐和镇痛反应，无相关不良反应。 [2] [8] [54] – [58] 两项对照临床研究比较了止吐、镇痛和抗分泌功效奥曲肽（300 µg/天）与丁溴东莨菪碱（60 mg/天）治疗 MBO 的比较。在这两项研究中，奥曲肽在所有反应参数（减少消化道分泌过多以及控制恶心和呕吐）方面的疗效均具有统计学意义。 [54] – [56] 一项 II 期研究表明，奥曲肽长效制剂（LAR Depot）与皮质类固醇联合治疗可有效且安全地治疗腹膜癌病引起的 MBO。 [59] 最近的文献综述得出结论，尽管对照临床试验的数量有限，但根据 15 项一致研究的结果和 20 项研究的经验，奥曲肽是治疗 MBO 的首选抗分泌药物。使用年限。 [60] 组胺-2 拮抗剂和质子泵抑制剂可用于减少胃分泌。最近的一项荟萃分析证实，雷尼替丁作为抗分泌剂比质子泵抑制剂更有效。基于这些数据，作者推测雷尼替丁可作为 MBO 抗分泌疗法的辅助药物，并建议开展具体研究来证实这些发现。 [61] [62]

The palliative treatment of MBO is polymodal and based on the combined use of different drugs active in controlling symptoms. According to most researchers and the recently published guidelines of clinical practice from the National Comprehensive Cancer Network,[63] the initial treatment for inoperable MBO is the combined use of analgesia with opioids, antiemetics, antisecretors, glucocorticoids, and hydratation with endovenous sera. It is reasonable to consider that the continuous infusion of fentanyl using an intravenous, subcutaneous route, or transdermic device may be the method of choice for its lesser influence on the intestinal motility and better tolerance in dehydrated patients. In complete MBO, the antiemetic of choice is haloperidol since the prokinetic effect of metoclopramide may paradoxically increase pain and nausea.[2] Antagonists of the 5-HT_3 receptors (ondansetron or Granisetron) may be an alternative for patients who have had an inadequate response to previous antiemetic treatments.[7] The initial use of glucocorticoids is recommended due to their antiemetic effect and reduction of intestinal edema, which may facilitate the spontaneous resolution of the occlusive picture.[32,50,51] Most researchers recommend the early use of octreotide or an antisecretor drug due to its clear superiority over other anticholinergic drugs.[2,54–59] Nasogastric aspiration should only be considered for the treatment of inoperable MBO in the absence of a symptomatic response to polymodal palliative treatment. The rate of control for nausea, vomiting, and pain using different variations of this polymodal palliative treatment strategy in inoperable MBO is greater than 80%, with spontaneous resolution in more than 30% of cases.[2,4,5,7,8,10,56] The estimated median survival in cases of inoperable MBO is 1 month with a 6-month life expectancy rate of less than 8%.[5]

MBO 的姑息治疗是多模式的，基于联合使用不同药物来有效控制症状。根据大多数研究人员和国家综合癌症网络最近发布的临床实践指南，[63] 无法手术的 MBO 的初始治疗是结合使用镇痛剂与阿片类药物、止吐剂、抗分泌剂、糖皮质激素以及水合药物静脉内血清。有理由认为，通过静脉、皮下途径或透皮装置持续输注芬太尼可能是首选方法，因为它对肠道蠕动的影响较小，脱水患者的耐受性较好。在完全性 MBO 中，首选止吐药是氟哌啶醇，因为甲氧氯普胺的促动力作用可能反而会增加疼痛和恶心。[2] 5-HT_3 受体拮抗剂（昂丹司琼或格拉司琼）可能是对先前止吐治疗反应不足的患者的替代药物。[7] 建议首先使用糖皮质激素，因为其具有止吐作用和减少肠水肿，这可能有助于闭塞图像的自发消退。[32,50,51] 大多数研究人员建议尽早使用奥曲肽或抗分泌药物，因为其明显优于其他抗胆碱能药物。[2,54–59] 只有在多模式姑息治疗没有症状反应的情况下，才应考虑鼻胃抽吸治疗无法手术的 MBO治疗。 在无法手术的 MBO 中，使用这种多模式姑息治疗策略的不同变体，恶心、呕吐和疼痛的控制率大于 80%，超过 30% 的病例可自行缓解。[2,4,5,7,8,10,56] 无法手术的 MBO 病例的估计中位生存期为 1 个月，6 个月预期寿命率低于 8%。[5]

Factors affecting the spontaneous resolution of inoperable MBO

影响无法手术的MBO自然消退的因素

The spontaneous resolution of the occlusive picture occurs in 30%–40% of patients with inoperable MBO.[5–9,64] Little is known about the factors that may influence the spontaneous resolution of this complication. Surgical studies describe patients undergoing surgery, but do not report the evolution of inoperable patients.

30%–40% 的无法手术的 MBO 患者会出现闭塞图像的自发消退。[5–9,64] 对于可能影响这种并发症自然消退的因素知之甚少。外科研究描述了接受手术的患者，但没有报告无法手术患者的进展情况。

A prospective cohort study was conducted by the Catalan Institute of Cancer in 2007, which included 100 patients diagnosed with inoperable MBO who were selected out of 885 patients visited by the Palliative Care Hospital Support Team (MBO prevalence = 11.3%). Twenty-five percent of these patients had previous episodes of MBO with spontaneous resolution (overall mean of 1.37 episodes per patient; SD ± 0.7; range 1–4). An extensive record of the clinical characteristics of these patients was documented. Over 80% of patients had multiple obstructive levels and more than 60% showed a peritoneal carcinomatosis with radiological or cytological confirmation. The spontaneous resolution of inoperable MBO with symptomatic treatment was observed in 42% of patients. Resolution occurred within the first 7 days after diagnosis in 92% of patients. In the follow-up procedure, the index of intestinal reobstruction was 74%. The mean overall survival rate was 23 days (95% CI = 16.8–29.4). Clinical data for all patients were stratified according to their specific evolution (spontaneous resolution versus no resolution) in order to indentify the factors influencing the spontaneous resolution of MBO. The mean survival was 12 days (95% CI = 9.0–14.1) for patients with no spontaneous resolution of MBO, and 57 days for patients with complete resolution ($P < 0.001$). In the group of patients who did not present MBO resolution, some showed tolerance to minimal food intake, mainly liquids, without recovery of normal digestive transit and with the need to maintain antiemetic and antisecretory treatment. The mean survival rate of these patients (persistent subobstruction) was 23 days (95% CI = 3.9–36), which is lower than the full resolution cases and higher than those patients who did not tolerate the intake of liquids at any time ($P < 0.001$). A multivariate analysis of the clinical characteristics of the patients assessed at the time of inclusion in the study reveals the most relevant factors influencing the consolidation and nonresolution of MBO. These are:

cognitive failure, cachexia, dyspnea at rest, palpable abdominal tumors, hepatic failure, upper intestinal obstruction, and dehydration.[5,64]

加泰罗尼亚癌症研究所于 2007 年进行了一项前瞻性队列研究，其中包括从姑息治疗医院支持团队访问的 885 名患者中选出的 100 名被诊断为无法手术的 MBO 患者（MBO 患病率 = 11.3%）。这些患者中 25% 的患者既往有过 MBO 发作并可自行缓解（每位患者总体平均发作 1.37 次；SD ± 0.7；范围 1-4）。对这些患者的临床特征进行了大量记录。超过 80% 的患者有多个梗阻水平，超过 60% 的患者经放射学或细胞学证实显示腹膜癌病。 42% 的患者经对症治疗后观察到无法手术的 MBO 自行消退。 92% 的患者在诊断后的前 7 天内得到缓解。随访过程中，肠再梗阻指数为74%。平均总生存率为 23 天（95% CI = 16.8–29.4）。所有患者的临床数据根据其具体进展（自发缓解与无缓解）进行分层，以确定影响 MBO 自发缓解的因素。 MBO 未自发消退的患者的平均生存期为 12 天（95% CI = 9.0-14.1），完全消退的患者的平均生存期为 57 天（P < 0.001）。在未出现 MBO 缓解的患者组中，一些患者表现出对最低限度食物摄入（主要是液体）的耐受性，无法恢复正常的消化转运，需要维持止吐和抗分泌治疗。这些患者（持续性亚梗阻）的平均生存率为 23 天（95% CI = 3.9-36），低于完全缓解病例，高于任何时候不能耐受液体摄入的患者（P < 0.001）。对纳入研究时评估的患者临床特征进行的多变量分析揭示了影响 MBO 巩固和未解决的最相关因素。这些是：认知障碍、恶病质、休息时呼吸困难、可触及的腹部肿瘤、肝功能衰竭、上肠梗阻和脱水。[5,64]

It is important to know the risk of nonresolution of MBO in order to carefully establish therapeutic measures, adjust real expectations, and accurately report them to the patient and family.

了解 MBO 未解决的风险非常重要，以便仔细制定治疗措施、调整实际期望并向患者和家属准确报告。

It is relevant to determine if there are pharmacological and nonpharmacological measures for preventing reobstruction in patients who demonstrate the spontaneous resolution of MBO. Some researchers suggest that a low residue diet, avoidance of osmotic laxatives, or use of longterm antisecretory drugs (eg, long-acting octreotide) may improve the likelihood of further obstructive episodes. However, this question remains unanswered. A pilot study conducted in 2005 including 15 ovarian cancer patients diagnosed with inoperable MBO, documented peritoneal carcinomatosis. These patients were treated with the immediate release octreotide and thereafter with long-acting octreotide administered monthly. Sixty percent of patients received at least one dose of long-acting octreotide. Twenty percent of the patients presented full recovery of digestive transit. These patients continued the antisecretory therapy with long-acting octreotide over a mean time of 9 months (3–15 months).[65] This long period of treatment with long acting octreotide, even higher to survival of many studies without this drug, may suggest that maintenance of antisecretory therapy may prevent new episodes of MBO. However, from data of this study it is not definitively possible to conclude that the long-acting octreotide is useful in preventing new episodes of the MBO, because patients were also treated with chemotherapy, which obviously influences the evolution of the MBO, and that new obstructive episodes were not reported clearly. At present the measures for preventing intestinal reobstruction remain under debate and may be the focus of future research.

确定是否有药物和非药物措施来预防 MBO 自发消退的患者再梗阻是相关的。一些研究人员建议，低残留饮食、避免渗透性泻药或使用长期抗分泌药物（例如长效奥曲肽）可能会增加进一步梗阻发作的可能性。然而，这个问题仍然没有答案。 2005 年进行的一项试点研究包括 15 名被诊断患有无法手术的 MBO 的卵巢癌患者，并记录了腹膜癌病。这些患者接受速释奥曲肽治疗，然后每月接受长效奥曲肽治疗。百分之六十的患者接受了至少一剂长效奥曲肽。百分之二十的患者消化运输完全恢复。这些患者继续接受长效奥曲肽抗分泌治疗平均时间为 9 个月（3-15 个月）。[65] 长效奥曲肽的长期治疗，甚至比许多不使用这种药物的研究的生存率更高，可能表明维持抗分泌治疗可能会预防新的 MBO 发作。然而，从本研究的数据来看，并不能明确得出长效奥曲肽对于预防新的 MBO 发作有用的结论，因为患者还接受了化疗，这显然会影响 MBO 的演变，并且新的 MBO 发作可能会发生。阻塞性发作没有明确报告。目前预防肠道再梗阻的措施仍存在争议，可能是未来研究的重点。

Summary of key points 要点总结

– MBO is a frequent complication in advanced oncological patients, especially in those with abdominal tumors. The prevalence and incidence of global MBO in cancer and different primary organs requires elucidation since most studies are retrospective and based on a series of cases or contain heterogeneous criteria for outcomes and diagnoses.

– MBO 是晚期肿瘤患者的常见并发症，尤其是腹部肿瘤患者。癌症和不同原发器官中全球 MBO 的患病率和发生率需要阐明，因为大多数研究都是回顾性的，基于一系列病例或包含不同的结果和诊断标准。

– The initial diagnosis of MBO is fundamentally based on anamnesis, physical examination, and simple radiology of the abdomen. Radiological techniques with contrast, CT, and MR may increase the diagnostic precision related to tumor extension and the level of obstruction, which is often necessary for decision making and evaluation related to the indication for surgery or endoscopic palliation.

– MBO 的初步诊断基本上基于病史、体格检查和简单的腹部放射学检查。对比剂、CT 和 MR 等放射学技术可以提高与肿瘤扩展和梗阻程度相关的诊断精度，这对于与手术或内镜姑息治疗指征相关的决策和评估通常是必要的。

– Surgery is the only therapeutic measure that may reestablish digestive transit and allow – according to the evolution of the patient – treatment with the intention of eradicating or palliating the obstruction by intestinal bypass. However, its indication should be assessed carefully on an individualized manner, especially in patients with advanced cancer due to the high rate of surgical mortality and morbidity. Most experts claim the factors limiting surgery in MBO are advanced age, previous malnutrition, and the presence of multiple occlusive levels, extra-abdominal metastatic disease, refractory ascites, deteriorated general status, previous abdominal radiotherapy, and the absence of active oncological treatments.

– 手术是唯一可以重建消化道的治疗措施，并且可以根据患者的病情进展进行治疗，以消除或减轻肠道旁路阻塞。然而，其适应症应以个体化方式仔细评估，特别是对于晚期癌症患者，因为手术死亡率和发病率较高。大多数专家声称，限制 MBO 手术的因素包括高龄、既往营养不良、存在多个闭塞水平、腹外转移性疾病、难治性腹水、全身状况恶化、既往腹部放疗以及缺乏积极的肿瘤治疗。

– Self-expanding duodenal and colonic stents are highly effective and safe alternatives in patients with a single level of occlusion and who are considered inoperable.

– 对于单级闭塞且被认为无法手术的患者，自膨式十二指肠和结肠支架是高效且安全的替代方案。

– Percutaneous gastrostomy allows for more comfortable and safe longterm digestive decompression than nasogastric aspiration in patients with inoperable MBO and symptomatology that is inadequately controlled by symptomatic treatment.

– 对于无法手术的 MBO 和对症治疗无法充分控制症状的患者，经皮胃造口术可以比鼻胃抽吸术提供更舒适、更安全的长期消化减压。

– Palliative medical treatment of inoperable MBO is polymodal and based on the combined use of glucocorticoids, antiemetics, antisecretors, and potent analgesic opioids. Due to their antiemetic action and reduction of perilesional edema, glucocorticoids are indicated in the initial phases of this complication and may increase the rate of spontaneous resolution. Most experts concur that the antiemetics of choice are neuroleptics (haloperidol) and rule out the use of prokinetic drugs because of the risk of increasing pain if the obstructive process is not functional. Antagonists of 5-HT_3 receptors are effective for controlling emesis in the treatment MBO, even in cases where the patient's response to antiemetics is insufficient. The limitation of these considerations is they are based on the opinion of experts or noncontrolled clinical studies (Phase II).

– 无法手术的 MBO 的姑息治疗是多模式的，基于糖皮质激素、止吐剂、抗分泌剂和强效镇痛阿片类药物的联合使用。由于糖皮质激素具有止吐作用和减轻病灶周围水肿的作用，因此在该并发症的初始阶段使用糖皮质激素，可能会提高自然消退的速度。大多数专家一致认为止吐药的选择是精神安定药（氟哌啶醇），并排除使用促动力药物，因为如果阻塞过程不起作用，则有增加疼痛的风险。 5-HT_3 受体拮抗剂可有效控制 MBO 治疗中的呕吐，即使在患者对止吐药反应不足的情况下也是如此。这些考虑因素的局限性在于它们基于专家或非对照临床研究（第二阶段）的意见。

– Abdominal pain in MBO is highly prevalent, of great intensity, and often requires the use of potent opioid drugs. No controlled clinical trials have compared the different potent opioids in this indication. Fentanyl is the opioid that least affects intestinal motility, which has been demonstrated by controlled clinical studies of different indications of MBO.

– MBO 中的腹痛非常普遍、剧烈，通常需要使用强效阿片类药物。尚无对照临床试验比较不同强效阿片类药物在该适应症中的作用。芬太尼是对肠道蠕动影响最小的阿片类药物，这一点已通过 MBO 不同适应症的对照临床研究得到证实。

– Antisecretor drugs improve nausea, vomiting, and pain with an important reduction in intestinal hypersecretion proximal to the occlusive process. According to the results of several controlled clinical studies, octreotide, an analog of somatostatin and a potent antisecretor drug, has shown a clearly superior efficacy with anticholinergic drugs. As mentioned previously, the implementation or maintenance of digestive aspiration with a nasogastric tube or percutaneous gastrostomy is only useful if the polymodal palliative pharmacological treatment cannot achieve adequate symptomatic control.

– 抗分泌药物可改善恶心、呕吐和疼痛，并显着减少闭塞过程附近的肠道分泌过多。根据多项对照临床研究的结果，奥曲肽（生长抑素的类似物和强效抗分泌药物）已显示出明显优于抗胆碱能药物的疗效。如前所述，只有当多模式姑息性药物治疗无法实现充分的症状控制时，通过鼻胃管或经皮胃造口术实施或维持消化道抽吸才有用。

– Symptomatic control is very high with the medical treatment strategy and spontaneous resolution is observed in more than one third of patients.

– 通过药物治疗策略，症状控制非常好，超过三分之一的患者观察到症状自行缓解。

– The life expectancy of advanced oncological patients from the diagnosis of inoperable MBO is short, with an estimated mean survival rate no longer than 4 weeks.

– 晚期肿瘤患者被诊断为无法手术的 MBO 后的预期寿命很短，估计平均生存率不超过 4 周。

– The most relevant factors influencing the consolidation and nonresolution of MBO include cognitive failure, cachexia, dyspnea at rest, palpable abdominal tumors, hepatic failure, upper intestinal obstruction, and dehydration.

– 影响 MBO 巩固和未解决的最相关因素包括认知障碍、恶病质、休息时呼吸困难、可触及的腹部肿瘤、肝功能衰竭、上肠梗阻和脱水。

Footnotes 脚注

Disclosure 披露

The authors declare no conflict of interest in this work.

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